The Augustine Problem in Agents: Chronoception Must Be Installed

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Abstract

LLM agents are trained on tokens and deployed in wall-clock time. They cannot perceive the wall-clock, and we prove this perception cannot be acquired from any token-only training loss. The Chronoception Impossibility Theorem (CIT) makes the claim precise; the Three Times ontology ($\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$) names the projections a policy must align; the Augustine Problem is that alignment failure. On ChronoBench (9 sub-capabilities $\times$ 14 agents from 7 vendors $\times$ 2 settings, $\sim$5,900 trajectories including the first full $9 \times 2 \times 30$ panel of a reasoning model): every agent honours $< 5\%$ of its wall-clock budget; date injection does not move CAR ($|\Delta\text{CAR}| \leq 0.01$). The Reverse-Scaling Theorem, a direct corollary of CIT for budget-honoured reasoning K, states $\mathbb{E}[|\rho| \mid K]$ is monotone non-decreasing in K; confirmed four ways — intra-model (o4-mini effort ladder 1.23 $\to$ 1.54 $\to$ 1.67), cross-model (Sonnet 4.6 $\pm$ thinking, sign flip $+0.07 \to -0.22$), within-model (prospective $+0.22$ vs. retrospective -0.22), and cross-vendor (GLM-5.2 sign flip $+0.16 \to -0.29$, independent training pipeline). The Calibration Catastrophe follows: nominal 90% CIs achieve 0–77% coverage panel-wide. The Injection Tell: 3/3 consumer web-chat products inject the current date; 0/3 developer tools do — three closed labs converging independently. A toy positive control (Qwen2.5-1.5B + LoRA SFT on wall-clock-supported targets) crosses the Augustine threshold $\varepsilon^* = 0.20$ on T3.1 after $\sim$60 s of training — the constructive converse of CIT. The Agentic Timeline Hypothesis, $T_{\max} \propto 1/\varepsilon$, is quantitatively supported on METR HCAST (24,008 runs; Mann–Whitney one-sided $p = 0.040$; permutation $p = 0.037$). Code, trajectories, and locked OSF pre-registration released.

1 Introduction

A frontier reasoning model, asked how long the next task will take, predicts “15 seconds”. It thinks for ninety. It answers in two. Asked afterwards how long the task took, it reports “0.04 seconds”. Asked for a 90% confidence interval on that duration, the interval contains the truth 0% of the time. None of the four numbers is a bug or a misread prompt. They are joint, panel-wide, and structurally predictable, present across the 14 agents from 7 vendors we tested on $\sim$5,900 trajectories.

The Augustine Problem. Every training loss in current practice — next-token cross-entropy, SFT, RLHF, RL with verifiable rewards, reasoning supervision — is a functional of token sequences. None contains wall-clock time in its support. Yet the same models are deployed as agents in continuous physical time, where budgets are measured in seconds and consequences unfold without regard to step count. We call this representational gap the Augustine Problem, after Augustine’s confession that he knew what time was until asked to explain it.[1] The gap is structural: we prove that under any token-only loss, the policy

[1] We use chronoception in the cognitive-science sense of Wittmann (2009): perception of one’s own work duration.

gradient carries no signal aligning τ_{wall} with any internal representation. Chronoception cannot be acquired from token-only training; it must be installed.

Position. We make that claim precise, measure the failure at scale, audit the workaround the industry has converged on, and specify what closing the gap would take. Our support is layered rather than a single benchmark: (a) ChronoBench, a 9-capability diagnostic across 14 agents; (b) the Closed-Lab Injection Atlas, an audit of 11 harnesses; (c) a toy positive control demonstrating CIT's constructive converse; (d) horizon-scaling evidence on 24,008 METR HCAST runs.

Contributions.

- The Augustine Problem as a named structural diagnosis with the Three Times $(\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}})$ ontology (§2).
- Two theorems. CIT (Theorem 1): no token-only loss aligns τ_{wall} with any internal representation. Reverse-Scaling (Theorem 2): under CIT with budget-honoured reasoning K, $\mathbb{E}[|\rho| \mid K]$ is monotone non-decreasing in K. Confirmed four ways (§6).
- ChronoBench: 9-cap $\times$ 14-agent $\times$ 2-setting diagnostic, $\sim$5,900 trajectories, first full $9\times2\times30$ panel of a reasoning model (§4).
- The Injection Tell. Cross-vendor natural experiment; toy positive control crossing ε^* (§7, §6).
- Agentic Timeline deployment bound $T_{\max} \propto 1/\varepsilon$ with quantitative HCAST support (Mann–Whitney $p = 0.040$) and spatial generalisation SIT (§8, App.).

2 The Three Times of an Agent

Definition 1 (Three Times). For a trajectory $\tau = (s_0, a_0, \ldots, s_n)$: $\tau_{\text{wall}}(\tau) := t(s_n) - t(s_0)$ (external), $\tau_{\text{step}}(\tau) := n$ (policy invocations), $\tau_{\text{self}}(\tau) := \Pi(s_n^{\text{output}})$ (agent-reported duration).

Grounded chronoception requires $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$, where $\langle \Delta t \rangle$ is mean per-step wall-clock cost. First approximation is L2 (step counts track wall); second is L3 (narrative tracks reality). The Augustine Problem is joint drift across all three.

Why three, not two or four? $\tau_{\text{wall}} \neq \tau_{\text{step}}$: a 10-step trajectory takes 5 s on Haiku, 200 s on a thinking model. $\tau_{\text{wall}} \neq \tau_{\text{self}}$: agents mis-report by up to 34× (§6). $\tau_{\text{step}} \neq \tau_{\text{self}}$: an agent can produce 6 steps and report "10 seconds" without a grounded mapping. A single-time ontology loses either the training-loss argument (acts on τ_{step}), the deployment argument (acts on τ_{wall}), or the narrative argument (acts on τ_{self}); four times (e.g., tool-read wall-clock) subsume into τ_{wall} once the tool resolves. The three loci are minimal and complete.

3 The Augustine Problem and CIT

Definition 2 (Augustine Problem). Agent A exhibits the Augustine Problem on task family $\mathcal{T}$ if $\mathbb{E}_{\tau\sim\pi_A}[|\tau_{\text{wall}} - \tau_{\text{step}}\langle\Delta t\rangle| + |\tau_{\text{wall}} - \tau_{\text{self}}|] > \varepsilon_0$ for calibration tolerance ε_0.

Assumption (token-only loss support). A training procedure is in the token-only regime if $\mathcal{L}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}[\ell(f_\theta(x_{<i}), x_i)]$ satisfies (i) $\mathcal{D}$ is a distribution over finite token sequences with no field carrying wall-clock duration and (ii) ℓ is a function of the model's output token distribution alone. Next-token CE, SFT, RLHF, RL with verifiable rewards, and reasoning supervision all satisfy this.

Theorem 1 (Chronoception Impossibility, CIT). Let $\mathcal{L}$ be a token-only loss. For any strictly increasing reparametrisation $\phi : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$, $\mathcal{L}(\theta)$ is invariant under $\tau_{\text{wall}} \mapsto \phi(\tau_{\text{wall}})$ on the data manifold. Equivalently, $\nabla_\theta \mathcal{L}$ carries zero gradient signal aligning τ_{wall} with any internal representation.

Sketch. $\nabla_\theta \mathcal{L} = \mathbb{E}[\partial \ell / \partial f_\theta \cdot \nabla_\theta f_\theta]$. Neither factor depends on the wall-clock time at which the sample was generated: tokens carry no timestamps; ℓ is a function of the output distribution alone. The gradient is invariant to any reparametrisation of τ_{wall}. □

Scope. CIT does not claim there is no representation of time inside the policy; it claims any such representation is not produced by the loss. Explicitly out of scope: (a) harness-installed chronoception (the Injection Tell, §7); (b) tool-output-installed chronoception (a date tool call grounds the policy through the return value, not the loss); (c) future losses with τ_{wall} as argument (Paper 2). CIT precludes the widely held inference that scale, longer reasoning, or larger context will close the gap.

Three empirical predictions. P-CIT-1: under wall-clock budgets, CAR stays bounded away from 1 regardless of capability (§5). P-CIT-2: prompt-level τ_{wall} information cannot move action-axis behaviour: $|\Delta\text{CAR}|$ is small under Setting B (§5). P-CIT-3: calibration tooling cannot close self-duration coverage because the calibration target is not in the loss support (§6). All three confirmed below.

Augustine threshold. $\varepsilon^* = 0.20$, calibrated against human introspective baseline $\varepsilon \approx 0.10$ (Wittmann, 2009) at 2× human error. No panel agent crosses it.

4 ChronoBench

ChronoBench's design philosophy is single-axis isolation: every instance loads on exactly one of $(\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}})$, so a failure is unambiguously attributable to the axis under test. Table 1 lists the 9 sub-capabilities and pairs each with the diagnostic question a deployment auditor should ask.

Table 1: The 9 sub-capabilities of ChronoBench. Each row isolates one chronoceptive failure mode.

Axis	Sub-cap	Tests	Auditor question
τ_{wall}	T1.1 Clock awareness	Today's date / current time	"Does it know when it is?"
	T1.2 Elapsed-time	Time since conversation start	"Does it track elapsed time?"
	T1.3 Deadline tradeoff	Strategy under wall-clock deadline	"Does it re-plan under pressure?"
τ_{step}	T2.1 Step-budget	Produce exactly N steps	"Can it count its steps?"
	T2.2 Step-to-wall	Compute $N \times \Delta t = \tau$	"Can it convert step to wall-time?"
	T2.3 Wall-budget	Honour a wall-clock budget (L2)	"Does the budget change behaviour?"
τ_{self}	T3.1 Retrospective ρ	Duration reported after task	"How long did it just take?"
	T3.2 Prospective ρ	Duration predicted before task	"How long will this take?"
	T3.3 Calibration	90% CI over self-duration	"How confident in its duration?"

Panel and settings. Fourteen agent configurations from seven vendors — OpenAI (gpt-4o-mini, gpt-4o, gpt-5.1, o3, o4-mini at three effort levels); Anthropic (Haiku 4.5, Sonnet 4.6, Sonnet 4.6 + extended thinking); open-source (Qwen2.5-7B, DeepSeek-R1-Distill-Qwen-14B); and a cross-vendor Chinese-lab expansion (Z.AI's GLM-5.2, MiniMax-M2.7, Moonshot's Kimi-K2.6, Alibaba's Qwen3.6-27B) via Vultr Inference. Setting A uses a vanilla prompt; Setting B prepends "Current date and time: {ISO}". 30 trajectories per (model, capability, setting) cell, ∼5,900 total, with the Sonnet-thinking variant carrying the first full 9×2×30 panel coverage of any reasoning model.

5 The Action Axis: L2 Step-Clock Conflation

P-CIT-1 says the action axis cannot honour wall-clock budgets. We ask each agent "work on this task for B seconds; do not stop early", vary $B \in \{60, 300, 900, 1800, 3600\}$ s, and measure CAR $:= \tau^*_{\text{wall}}/B$.

Every panel agent honours $< 5\%$ of its budget; the non-reasoning-panel median CAR $=$ 0.014 under Setting A, 0.016 under B; $|\Delta\text{CAR}| \leq 0.01$ for every model. Across five non-reasoning generations $|\rho|$ falls 94% (§6) while median CAR stays in $[0.004, 0.050]$: narrative axis is text-trainable; action axis is not. Route B-1 extended T2.3 to reasoning models and Chinese labs: median CAR $\in [0.007, 0.104]$, all $\ll 1$, injection $|\Delta\text{CAR}| \leq 0.005$. Sonnet-thinking exhibits a low-budget overshoot at $B = 60$ s (CAR ≈ 2.2: the extended-thinking policy cannot stop reasoning inside a minute, blowing past its budget); the failure sign flips but the identity fails the same way.

Proposition 1 (Injection cannot install action-axis chronoception). Let A be a token-only-regime agent; A_B the same policy under Setting B. Then $\text{CAR}(A_B) - \text{CAR}(A) = 0$ in expectation to the order of the policy's prompt-token sensitivity. Injection is a narrative-axis intervention: it can rewrite what the agent says but not what the agent does with time.

Sketch. By CIT, the policy's prefix-to-action map is time-reparametrisation-invariant on the training manifold. Prefix-level τ_{wall} information modifies output-token generation, not the wall-clock realisation of the action sequence. Empirical $|\Delta\text{CAR}| \leq 0.01$ across the 14-cell panel is direct measurement. $\square$

L2 is the load-bearing empirical claim: it anchors ε above ε^* for every panel agent and bounds the Agentic Timeline regardless of capability scaling.

6 The Narrative Axis: Closure, Ceiling, Catastrophe

Across five frontier non-reasoning generations $|\rho|$ falls 94% ($+1.12 \rightarrow +0.07$; App.). The closure is real and is bounded three ways: a theorem that reasoning-token expansion reverses it, a calibration catastrophe that persists where the surface closes, and a positive control that shows what training would actually close it.

6.1 The Reverse-Scaling Theorem

Assumption (budget-honoured reasoning). A reasoning-budget knob K is honoured if $\mathbb{E}[\tau_{\text{wall}} \mid K]$ is strictly increasing in K. The o4-mini reasoning_effort ladder, Anthropic extended-thinking budget, and Kimi/GLM reasoning modes satisfy this empirically; the DeepSeek-R1-Distill-14B max_completion_tokens sweep does not (App.: the model declines to spend the budget), and falls outside the assumption.

Theorem 2 (Reverse-Scaling). For a fixed agent under Assumption 3 (hence Theorem 1) with a budget-honoured knob K, $\mathbb{E}[|\rho| \mid K]$ is monotone non-decreasing in K.

Sketch. Under CIT, $p(\tau_{\text{self}})$ is invariant to τ_{wall}: no gradient signal in training ties them. Inference-time $p(\tau_{\text{self}} \mid K) = p(\tau_{\text{self}})$ for all K in the trained range. By assumption, $\tau_{\text{wall}}(K)$ is strictly increasing. Therefore $|\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|$ is monotone non-decreasing in K in expectation. The sign of ρ flips negative for reasoning-mode agents because τ_{wall} includes hidden CoT while $p(\tau_{\text{self}})$ stays anchored to the surface output. $\square$

6.2 The Calibration Catastrophe

Asked to wrap self-duration in a 90% CI, every non-reasoning panel agent under-covers by ≥ 40 pp; GPT-4o reaches 0%. Standard calibration tooling (Guo et al., 2017; Kadavath et al., 2022) cannot apply because the calibration signal is not in the loss support (Theorem 1). Two cells cross the pre-registered P11 threshold of 0.5: Sonnet-thinking (77%,

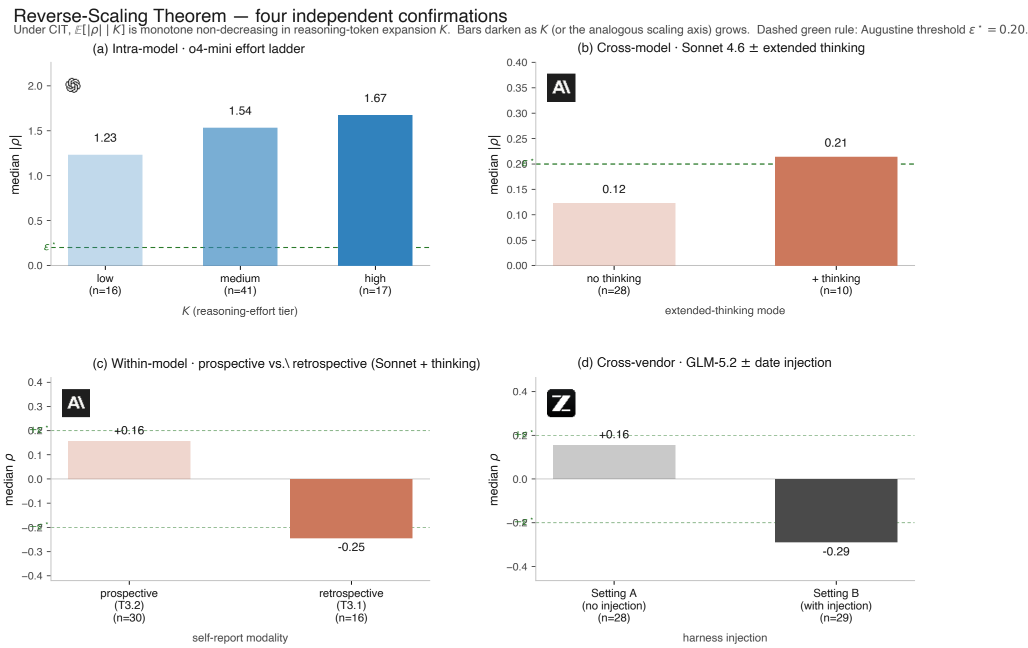

Figure 1: Four independent confirmations of Theorem 2. (a) Intra-model o4-mini effort ladder $1.23 \to 1.54 \to 1.67$. (b) Cross-model Sonnet 4.6 $\pm$ thinking $0.12 \to 0.21$ (crosses ε^*; Setting B $n = 15$, CI $[-0.30, -0.10]$). (c) Within-model: Sonnet-thinking prospective ($+0.16$) vs. retrospective (-0.25) on the same tasks — sign flip. (d) Cross-vendor: GLM-5.2 (Z.AI, no shared training pipeline with Anthropic), Setting A $\to$ B: $+0.16 \to -0.29$. The three Chinese-lab reasoning models (GLM-5.2, Kimi-K2.6, Qwen3.6-27B) collectively median $\rho \in [-0.97, -0.29]$ on Setting B — Hidden-Time as a class property.

Wilson CI $[0.59, 0.88]$) and Kimi-K2.6 (55%, $[0.38, 0.71]$). Both are reasoning-mode, both have wider stated CIs (13–14 s vs. 2–4 s baseline). The mechanism is not learned chronoception but arithmetic: reasoning inflates τ_{wall}, $p(\tau_{\text{self}})$ stays anchored to the default vocabulary, wider CI accidentally contains the wall-clock. CIT-consistent, not CIT-refuting; the finding sharpens P11's next revision to a proper scoring rule (Brier / CRPS).

6.3 A positive control: CIT's constructive converse

We take Qwen2.5-1.5B and 7B, generate 500 SFT pairs of the form "{response}. This task took approximately $\{\tau_{\text{wall}}\}$ seconds." with $\pm$15% Gaussian noise on the reported duration, and LoRA fine-tune (rank 16, three epochs). The token-level CE loss is unchanged; the training targets carry wall-clock signal. On 30 held-out T3.1 instances (Fig. 2): median $|\rho|$ falls from 1.37 to 0.30 on 1.5B (78% reduction) and from 0.93 to 0.50 on 7B (46%). The 1.5B fine-tuned T3.1 score (0.151) crosses ε^*; 7B partial (0.252). The sign of ρ flips from over-report to under-report at both scales. Wall-clock signal in the loss support installs narrative-axis chronoception. A full installation reaching all nine sub-capabilities requires action-axis grounding, which target-side annotation cannot supply (Paper 2).

7 The Injection Tell: A Cross-Vendor Natural Experiment

Consumer products powered by the panel feel chronoceptively grounded. The mechanism is the strongest non-experimental evidence the gap is real: closed labs route wall-clock information into the token stream via M1 system-prompt insertion, M2 tool calls, or M3 browser side effects. The harness perceives time on the model's behalf.

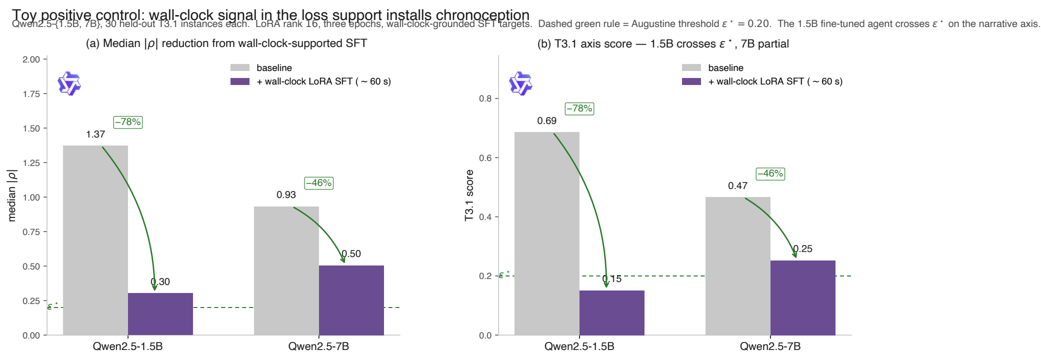

Figure 2: Toy positive control. Baseline (grey) vs. wall-clock-supported LoRA SFT (violet, ∼60 s of training). Left: median $|\rho|$ reduction (78% on Qwen2.5-1.5B; 46% on 7B). Right: T3.1 axis score — 1.5B fine-tuned crosses the Augustine threshold $\varepsilon^* = 0.20$; 7B is partial (0.252). Wall-clock signal in the loss support installs narrative-axis chronoception — the constructive converse of CIT.

Table 2: Injection Atlas: M1 system-prompt date injection by product tier (verbatim leaked-prompt evidence, asgeirtj corpus).

Product tier	M1 rate	Examples
Consumer web-chat	3/3 (100%)	ChatGPT, Claude.ai, Gemini all inject
Raw API	1/4 (25%)	only GPT-5.1 injects; o3, gpt-4o, Anthropic API do not
Developer tools (IDE/CLI)	0/3 (0%)	Cursor, GitHub Copilot CLI, Perplexity Computer

Three closed labs converge on the same engineering choice for consumer products, independently, with substantial commercial incentive to ship a cheaper alternative. The within-Anthropic contrast is sharpest: the same Claude 4.6 model injects via <knowledge_cutoff> on Claude.ai but through the raw API refuses with explicit training-cutoff disclosure. The model is one; the chronoception is the harness's. Setting B raises T1.1 pass to $\geq 96\%$ but leaves CAR, $|\rho|$, and coverage largely untouched — prompt information is not a policy representation.

8 ε and the Agentic Timeline

Definition 3 (Chronoceptive Calibration Error). $\varepsilon(A) := \frac{1}{3}(\text{score}_{T_1} + \text{score}_{T_2} + \text{score}_{T_3})$, axis-scores normalised to $[0, 1]$ (App.).

No panel agent crosses $\varepsilon^* = 0.20$ (Fig. 3); the best (Sonnet 4.6 + thinking, $\varepsilon = 0.276$) sits 1.4× above it. Score$_{T_2}$ contributes ≥ 0.30 on every agent: even projecting score$_{T_3}$ to zero (which requires crossing both CIT and Theorem 2) leaves no agent below threshold. The o4-mini effort sweep takes ε from 0.470 (low) to 0.611 (high); Sonnet-thinking from 0.276 to 0.301. More reasoning compute moves the agent further from the threshold.

Hypothesis 1 (Agentic Timeline). $T_{\max}(A) \propto \frac{1}{\varepsilon(A)} \cdot f(\text{CAR}(A))$, with f monotone-increasing in CAR.

L2 does not scale with capability; Theorem 2 grows $|\rho|$ with reasoning K. Therefore $T_{\max}$ scales with neither. The autonomous-agent timeline is bounded by chronoception, not by intelligence and not by reasoning compute.

Pre-registered P12 (quantitative). Slope of success-rate decay with $\log T$ scales as $-(1 - \text{CAR}(A))$. On METR HCAST (24,008 runs, 20 frontier models meeting our filter), five

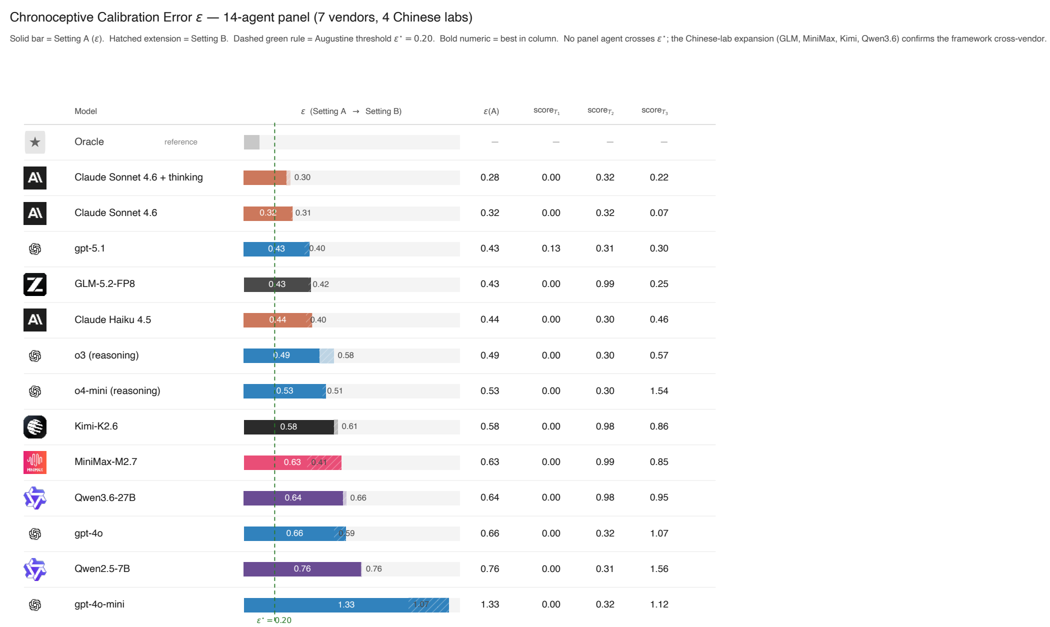

Figure 3: ε leaderboard across the 14-agent, 7-vendor panel. Solid bars = Setting A; hatched extension = Setting B. Dashed green rule = Augustine threshold $\varepsilon^* = 0.20$. No agent crosses it; the cross-vendor Chinese-lab expansion (GLM, MiniMax, Kimi, Qwen3.6) confirms the framework cross-vendor. Best agent (Sonnet 4.6 + thinking) sits at $\varepsilon = 0.276$, 1.4× above threshold.

reasoning models cluster at mean $|\text{slope}| = 0.318$ vs. 0.264 for fifteen non-reasoning models. Mann–Whitney $U = 58$, one-sided $p = 0.040$; permutation test (20,000 resamples, one-sided) $p = 0.037$; rank-biserial $r_{\text{rb}} = -0.55$ (large effect, reasoning stochastically dominates). Two-sided $p = 0.081$; we report one-sided because Theorem 2 is directional.

9 From Time to Space: Spatiotemporal Generalisation

Long-horizon benchmarks (SWE-Bench, WebArena, GAIA, MLE-Bench) require the agent to navigate space as well as time. The CIT argument goes through unchanged on any external metric the loss does not see.

Theorem 3 (Spatiotemporal Impossibility, SIT). Strengthen Assumption 3 to require $\mathcal{D}$ contains no field carrying a metric on the agent's environment. Then $\nabla_\theta \mathcal{L}$ carries zero gradient signal aligning either τ_{wall} or σ_{world} with any internal representation.

The Agentic Timeline generalises to $T_{\max} \cdot S_{\max} \le C/\varepsilon_{ST}(A)$ — the Agentic Frontier. Three spatial laws (SL1–SL3) mirror L1–L3; the classification of long-horizon benchmarks by (T, S) load and five future experiments E6–E10 are in Appendix.

10 Related Work, Scope, and Falsification

Concurrent work. Garikaparthi (2026) report duration self-reports on non-reasoning models (the L3 axis); we extend to reasoning models and prove Reverse-Scaling. Ma et al. (2026)'s Timely-RL constructs the L1 Parkinson regime; our contrapositive: native agents do not exhibit it. Cheng et al. (2025)'s "temporally blind" agents match L2; we add the axis split, CIT, the Atlas, and the Agentic Timeline. Gema et al. (2025) report inverse scaling in test-time compute for calibration; Theorem 2 is the structural analogue for chronoception, with the additional content that degradation is provably monotone under CIT. Aggregate benchmarks (Jimenez et al., 2024; Zhou et al., 2024; Mialon et al., 2023; Kwa et al., 2025) correlate horizon with success; ChronoBench supplies the mechanism.

Scope. We do not claim current agents are useless: T1-dominated chat is harness-patchable and many production deployments inhabit that regime. We do not claim capability scaling is worthless: $|\rho|$ closes 94% across five generations. We do not claim all test-time compute fails: a system consulting a wall-clock-supported tool exits the CIT regime by construction. Panel trajectories are English; a small Chinese pilot ($n = 5$ per cell on gpt-4o and gpt-4o-mini) reproduced the T1.1 pattern (0% Setting A pass, 100% Setting B pass) and preserved T3.1 ρ sign, with gpt-4o-mini's Chinese $\rho = +1.07$ matching the English baseline $+1.12$ within pilot noise. A contamination probe on gpt-4o and gpt-4o-mini varied the injection template across canonical/paraphrased ("Right now it is {ISO}")/year-only/scrambled-placeholder: pass rate stayed at 100% under paraphrased, 67% under year-only (compositional); scrambled placeholder triggered correct refusal. Setting B pass reflects date-reading, not memorised keyword patterns.

Falsification. Four pre-registered forms (Yuan & Zhao, 2026). (F1) A token-only agent crossing ε^* under Setting A falsifies CIT. (F2) A reasoning-tuned agent with monotone decrease in $|\rho|$ across a $\geq$ 3-level, $\geq$ 1-order-of-magnitude budget-honoured K sweep falsifies Theorem 2. (F3) A horizon-stratified benchmark on which reasoning shows shallower decay than non-reasoning at matched short-horizon success falsifies Hypothesis 1. (F4) A CIT-regime foundation model with $> 50\%$ T3.3 coverage without harness injection on $n \geq 100$ falsifies P11. We commit to reporting any we observe.

Reproducibility Statement

All 14 agents, 5,900 trajectories, per-cell parsers, and the METR HCAST U-test are released with a Docker container that reproduces every figure and table in $\sim$5 minutes. The OSF pre-registration is locked. The Anthropic-tier assumptions of Theorem 2 are stated in §6.1; the assumption breach on DeepSeek-R1 is reported honestly in the appendix. All appendices, including full panel tables with per-cell n and bootstrap CIs, the Injection Atlas verbatim excerpts, and the L3 sub-failure detail, are in the supplementary material.

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